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Equity principles: Using social theory for more effective social transformation in agricultural research for development

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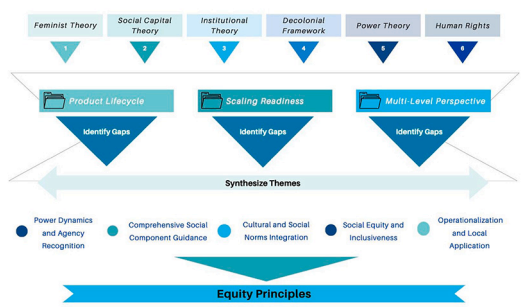
HIGHLIGHTS

- Agricultural Innovation Systems contribute significantly to achieving Sustainable Development Goals.
- Conceptualization of innovation systems and practical implementation have not historically incorporated social differences.
- Considering social variances, power dynamics, and agency reduces unintended consequences and leads to desired social goals.
- Equity Principles are created through applying social theory to address gaps in innovation systems and scaling tools.
- The Equity Principles can assist a wide range of actors incorporate diverse perspectives into agricultural innovation.

GRAPHICAL ABSTRACT



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ABSTRACT

CONTEXT: Agricultural innovations and their applications are increasingly recognized as crucial mechanisms for achieving the 2030 Sustainable Development Goals (SDGs). Actors in agricultural research for development (AR4D) frequently use Agricultural Innovation Systems (AIS) frameworks to comprehend the ecosystems within which innovations are developed and scaled. Given the SDGs' emphasis on social outcomes, a reflection on social diversity, power, and the integration of social theory into AIS and AR4D tools is crucial for addressing the nuances of social objectives.

OBJECTIVE: This research critically evaluates AIS frameworks and AR4D tools through applying social theory to enhance social outcomes. We offer practical application through the development of "Equity Principles for Social Transformation (EPs)." These EPs are designed to guide AR4D organizations in innovation and scaling efforts that

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effectively achieve meaningful social outcomes. Through this approach, we aim to enrich the conceptual understanding of equity within AIS and provide practical strategies for implementing these insights, thus empowering AR4D actors to be more effective.

METHODS: We start by selecting key social theories to analyze global power imbalances and local social exclusion within AIS frameworks and AR4D tools. Using these theories, we examine three case studies to uncover gaps in their approach to social dimensions. We categorize these gaps through thematic analysis and formulate EPs informed by social theories and a practical understanding of AR4D tools.

RESULTS AND CONCLUSIONS: Equity analysis of each case study reveals gaps in understanding social implications within upstream and downstream research efforts. These gaps include insufficient addressing of power dynamics and agency recognition, lack of comprehensive guidance on critical social components, oversight of cultural and institutional norms, exacerbation of social inequities, and the case studies' limitations in flexibility for addressing social inclusion effectively. Additionally, there is a notable lack of clear operational guidelines for applying the frameworks in diverse contexts, including the challenge of translating conceptual levels into local action. Seven EPs were developed: recognize AR4D power dynamics; define goals, anti-goals, and for whom; build global "horizontal" partnerships; acknowledge social differences among innovation users and non-users; innovate and curate innovation appropriately; assess impact and reflect; and develop systems capacity.

SIGNIFICANCE: The EPs connect innovation systems with positive social change. They help AR4D professionals consider and evaluate the impact of innovation. The EPs provide an additional framework that enables AR4D practitioners to prioritize user needs from the beginning, challenge biases, and more effectively achieve the social objectives outlined in the SDGs.

1. Introduction

Despite significant investments and global efforts spanning decades, the challenge to alleviate poverty and improve the lives of the most disadvantaged remains daunting. According to the World Bank (2018), 700 million people live on less than $1.90 a day; in Sub-Saharan Africa, one in three children suffer from stunted growth. Some 48% of those in poverty are based in rural locales, largely relying on small-scale agriculture (FAO, 2019). Investing in farmers and their communities is crucial for reducing poverty in rural and peri-urban areas. In the past 50 years, over $60 billion has been invested in agricultural interventions and innovations to address hunger and malnutrition challenges (Alston et al., 2020). As we approach the 2030 deadline for achieving the Sustainable Development Goals (SDGs), it is essential to critically address and reform the systemic issues that continue to place global populations in vulnerable positions. Such social transformation includes significant shifts or changes in individuals' and society's structure, functioning, and cultural norms. To achieve meaningful change, agricultural research for development (AR4D) actors are increasingly employing innovation strategies, or downscaling inefficient innovations, to realize international development goals (Goulet and Vinck, 2012; Hebinck et al., 2022; Kok and Klerkx, 2023; Woltering et al., 2024). Proposed solutions must address the complex and interlinked social, economic, cultural, and environmental factors that shape the lives of agriculture's most vulnerable actors, including small-scale farming households and their communities.

AR4D innovation creation and use is multidimensional and often mission-oriented (Herrero et al., 2020; Leeuwis and Aarts, 2021; Klerkx and Begemann, 2020). These interrelated actors and functions are often represented by agricultural innovation systems (AIS) frameworks that consider economic and political processes and temporal and relational interdependencies (Klerkx et al., 2012). They are conceptualized in many ways with varying degrees of complexity (Leeuwis et al., 2021; Wigboldus et al., 2016; Wigboldus and Leeuwis, 2013) to achieve different types of systems transformation (Weber et al., 2020; Klerkx and Begemann, 2020). AR4D actors leverage these frameworks to drive innovation and scale effectively, using them as both theoretical models and to guide the creation of AR4D tools to inform the direction of their innovation and scaling strategies.

Efforts have intensified to refine and operationalize AIS to better capture heterogeneity across geographical and cultural contexts and include ethical considerations (Jacobs et al., 2018; Kabirigi et al., 2022). However, research increasingly highlights unintended social and environmental consequences due to AR4D innovation and scaling (Ludwig and Macnaghten, 2020; McGuire et al., 2022), as well as non-diverse stakeholder representation in AR4D decision-making (Douthwaite et al., 2017). As AR4D actors try to reduce resource inequity and increase gender and social empowerment through innovation, they can inadvertently widen social disparities, hindering progress toward SDG achievement (McGuire et al., 2023).

Against this backdrop, this paper critically assesses prevalent AIS frameworks and AR4D tools, examining their alignment with social theory aimed at social transformation and overlooked social dimensions. We present key considerations and concepts, social transformation challenges and unintended consequences, relevant social theories that can address such issues, and areas of entry and reflection for social inclusion. We conduct a critical social analysis of prevalent AIS frameworks and AR4D tools: the Multi-Level Perspective (MLP) Framework, the Product Life Cycle (PLC) Approach from the United States Agency for International Development (USAID), and Scaling Readiness from the CGIAR, formerly known as the Consultative Group for International Agricultural Research. Insights from the analysis were then used to propose a set of *Equity Principles for Social Transformation* (EPs) that can be applied to AIS frameworks and AR4D tools, given that the practical learnings built from our analysis are also applicable to other frameworks. Our overarching goal is to offer these critiques and EPs to AR4D actors to spur more effective innovation and scaling aligned with the SDGs' aspirations for social transformation.

To do this, we set out to answer the following research questions:

1. Do AIS frameworks and AR4D tools give sufficient attention to social theory, and if not, how can traditional social theory methods address frequent oversights?
2. Given common emerging themes, can we establish standard principles that AR4D actors can universally adopt, irrespective of the AIS framework or AR4D tool they use?

The remainder of the paper is structured as follows: we first discuss key considerations for social transformation in AIS and provide an overview of the literature on selected pathways for addressing ethical challenges in innovation; we then present methods and summary of the results of the social analysis of AIS frameworks; subsequently, we present the proposed EPs; finally, we conclude with the discussion and conclusions of our findings. The appendix provides further details of the social analysis of the selected AIS frameworks/tools.

2. Applying social theory: Definitions and framing

2.1. Social transformation: A paradigm rooted in inclusivity

Social transformation refers to changes that occur within the structure of a society (Valladares, 2021) and can have different directionalities or missions, depending on the actors that define what that change should be (Hekkert et al., 2020; Klerkx and Begemann, 2020). This paper focuses on the SDG 2030 goals, which aim to 'leave no one behind' (Gupta and Vegelin, 2023; United Nations Committee for Development Policy, 2018). Achieving social transformation requires a holistic approach that integrates the social, economic, environmental, and cultural dimensions of communities and other living beings at multiple scales, as well as recognizing exnovation processes. As such, one-dimensional emphases on innovations are insufficient to reach all populations and even less so for vulnerable populations (Woltering et al., 2024). Ideally, social transformation extends beyond technological innovation to reshape how people interact with one another and their environment, emphasizing the interconnectedness of social, economic, and ecological well-being. This approach necessitates complementing natural science considerations with social science insights to comprehensively define the scope of 'all'.

Recognizing the interdependence of mechanisms that can lever social transformation, we highlight the need for innovations that address multiple SDGs simultaneously (Dörgo et al., 2018), such as SDG 1 (no poverty), SDG 2 (no hunger), SDG 5 (gender equality), SDG 10 (reduced inequality), and SDG 13 (climate action). For instance, social transformations in agriculture can enhance economic opportunities and knowledge for vulnerable farmers and their communities through a range of activities, including the establishment of strong farmer-led organizations, education, and the promotion of accessible sustainable farming practices. These practices not only increase crop yields but also protect the environment for the benefit of all living beings (Dixson-Decleve et al., 2022; FAO, 2017; Wood et al., 2023). Further, to achieve these comprehensive goals, it is essential to uplift vulnerable populations by developing and scaling innovations that favor these groups disproportionately (Dörgo et al., 2018).

This study analyzes what innovative or disruptive social considerations need to be introduced when planning, developing, and scaling innovations. Developing these innovations can internalize social transformation goals by innovating and establishing a systems approach fostering inclusive interactions among people and the environment in research processes. We explicitly propose including social considerations when developing innovations as a transformation pathway to support SDGs. Furthermore, while social transformation is an overarching aim of AR4D, in line with the Reach, Benefit, Empower, and Transform Framework developed by CGIAR's International Food Policy and Research Institute (Quisumbing et al., 2019; Quisumbing et al., 2023), social transformation remains aspirational within a continuum of positive (if not ideal) social change. In this paper, while the aim is social transformation, we recognize and discuss 'positive social outcomes' as beneficial outcomes as well.

2.2. Inclusive innovation as the catalyst for social transformation

We believe that the key to effective innovation for social transformation is inclusivity. Hoffecker (2021) argues that, in agriculture, innovation is deemed inclusive when it benefits the underprivileged both in the process and the outcomes. The CGIAR study supports the idea that a truly inclusive innovation considers factors such as product affordability, local entrepreneurship, capacity building, social empowerment, and systemic changes (Independent Science for Development Council, 2022). Examining scaling frameworks to assess their inclusivity, Sánchez Rodríguez et al. (2021) found that many frameworks tend to identify similar processes around planning, implementing, and adapting. By acknowledging inclusivity, the authors stress the need for increased dynamic and systems approaches when developing innovations. As we evaluate AIS frameworks and AR4D tools' potential effectiveness in envisioning and prioritizing mechanisms that lead to social transformation, we consider their process of inclusivity through responsible research and scaling, diversity of actors, and anticipation of negative consequences (Burget et al., 2017).

2.3. A prevalent challenge: Addressing unintended consequences of AR4D innovations

While AR4D actors strive to achieve positive social transformation, gender-responsive studies continue to illuminate unintended consequences when innovations and scaling processes inadequately account for marginalized populations (Gartrell, 1977; Paris and Chi, 2005; Cohen et al., 2016; Petesch et al., 2018; Quisumbing et al., 2019; Rola-Rubzen et al., 2020: Hackfort, 2021; Assan, 2023; Aghion and Griffith, 2022). By highlighting these persistent challenges, we offer an empirical and practical illustration of the substantial value social theory could add to addressing the complexities marginalized populations face. For instance, innovations often cater predominantly to able-bodied men, inadvertently neglecting the unique needs of women and other marginalized groups. This exclusion often manifests in tools or practices that lack cultural and physical appropriateness for diverse users. For instance, a well-intentioned innovation in Bolivia—a potato-sorting machine meant to reduce women's labor—was too cumbersome for women's use. This tool primarily benefited men, ironically exacerbating the gender disparity (de Roo et al., 2019; Polar et al., 2017).

As important as they are for disseminating agricultural knowledge, traditional communication mechanisms also persistently fail to cater to women's needs. Historical gender biases still stifle agricultural productivity and skew the transmission of essential information. As the agricultural sector increasingly integrates information and communication technologies for knowledge dissemination, this gendered rift deepens, leaving many women on the margins of technological advancements (Norton and Alwang, 2020; World Bank, 2017).

Women also consistently confront disparities in accessing vital resources, from land and labor to information and education. Such differential access, highlighted by studies like Ragasa (2012), underscores women's low technology adoption rates, exacerbating existing community inequities (de Roo et al., 2019). Additionally, financial mechanisms aiming to scale innovations often skew toward profit-centric models, frequently sidelining the intricacies required for accommodating vulnerable groups (Rietveld and van der Burg, 2021). Clear evidence of such oversight comes from Botswana, where men predominantly reaped the rewards of agribusiness training (Hovorka, 2005). Further, the process of technological adoption is nuanced, demanding alterations to established systems, reshaping interpersonal dynamics, and fostering new competencies. However, this adaptive process often sidelines those with limited community power, causing them to cede control over innovations (Mudege et al., 2020).

An increasingly established body of research highlights negative impacts on women and marginalized communities when innovation is introduced and scaled within a community (Galiè et al., 2017; McGuire et al., 2022). These instances indicate a need to re-think how upstream AR4D actors and downstream implementation processes relate within AIS frameworks and how to accommodate these dynamics in AIS conceptualization and AR4D tools.

2.4. Roles and relationships - AR4D actors in AIS

While recognizing the imperative of social transformation and its intricate ties to achieving the SDGs, it is essential to emphasize the mechanisms and actors that operationalize these changes within the agricultural sector. Central to this are AIS, which act as the bedrock upon which innovations for positive social change are built and disseminated. Since AIS are socio-technical systems, analysis requires both a social and

technological element for social transformation. AIS caters to the technical facets of innovation and the socio-political nuances that determine the success of an innovation in real-world contexts (Kok and Klerkx, 2023). By thinking through the functions, roles, and interactions of AR4D actors within AIS, we can understand the system's agency or social elements. We go beyond analyzing the system's technical and innovation-driven elements and see the relationships between the object (innovations) and the subject (actors) of the systems. In this way, we can better understand how different stakeholders, who have varying levels of influence, work together to shape the direction of innovations to align with social transformation goals.

AIS perspectives emphasize interactions and relationships among its system elements, including various innovations and stakeholders. This is crucial for facilitating knowledge generation, exchange, and application in agricultural innovation processes (Kabirigi et al., 2022). In turn, AIS frameworks help facilitate collaboration among a dynamic network of organizations and stakeholders to produce successful innovations (Aerni et al., 2015; Pound and Conroy, 2017). In the context of AR4D, actors and functions can be classified based on their participation in the innovation process's early (upstream) or later (downstream) stages. To understand and enhance the contributions of both upstream and downstream actors, specifically in terms of how they can better deliver innovations that lead to social transformation, we delineate three groups:

- **Upstream AR4D actors** include funders such as government agencies, national agriculture research services, private foundations, and researchers within large research entities and universities. The primary function of this group of actors are as decision-makers of AR4D, setting priorities for AIS development, engaging in research for innovation, and strategizing the scaling of innovation.
- **Downstream AR4D actors** include community-based organizations, national agriculture research services, agricultural extension agencies, and farmer community leaders. The primary function of this group of actors is to implement activities that scale innovations and ensure they reach the intended beneficiaries.
- **Intended users of AR4D innovation** include other relevant farmers and/or innovation users. The primary function of this group of actors is to use, test, and ultimately benefit from the innovation. The innovation will also impact a group of actors that are non-users.

While both upstream and downstream AR4D actors have traditional functions and roles within an AIS, they are not strictly confined to those roles. AR4D actors can engage in and benefit from both upstream and downstream processes. For example, the development phase of an innovation might involve active participation from its intended users, while scaling activities might be directly influenced by governmental bodies. Upstream actors can be considered beneficiaries within AR4D AIS as they enhance their own capacity and careers. Importantly, the primary objective remains to achieve positive development outcomes, with social transformation being our main concern.

Power dynamics are central to understanding the interplay and challenges within AIS. Upstream AR4D actors, given their positions, inherently possess more decision-making power and resources than their downstream counterparts. Funders have more power and agency than researchers due to their control over resources. Similarly, farmers often have less leverage over resources than researchers due to factors such as established social networks, management structures, levels of education, and available capital.

Upstream AR4D actors have primarily focused on inequity and diversity within the practical application of innovations 'in the field' while externalizing and compartmentalizing structural factors that contribute to social disparities in the landscapes where the innovations are envisioned, developed, and chosen. This becomes especially salient when the key decision-makers are not representative of the populations they intend to serve. Geographical dislocation, with many key AR4D actors rooted in high-income countries, can further skew perspectives and interests. This, coupled with systemic biases, such as those favoring natural sciences over social sciences, and societal issues, such as the sexism and racism pervasive in STEM (Science, Technology, Engineering, and Mathematics) fields, highlights the complexity of the AR4D landscape (Sokolova et al., 2023).

We posit that these intricate power dynamics and inherent biases can shape how AR4D actors innovate. Such dynamics introduce barriers to the equitable distribution of innovations and limit the broader transformative potential of AR4D initiatives. As such, the innovation process is *as vital* as the outcome it is intrinsically linked to. Therefore, applying social theory within this framework may assist in understanding the power dynamics and agency of upstream AR4D actors within AIS and their impact on downstream outcomes.

3. Methods

3.1. Aim and research questions

We aim to analyze the ways in which AIS frameworks and AR4D tools can theoretically and practically include established social theory to improve AR4D outcomes.

Question 1: Do AIS frameworks and tools give sufficient attention to social theory, and if not, how can social theory methods address frequent oversights?

Question 2: Given common emerging themes, can we establish standard principles that AR4D actors can adopt within the AIS frameworks or AR4D tools they use?

A critical social analysis was conducted to answer these questions, using three case studies consisting of an AIS framework and two AR4D tools: the Multi-Level Perspective Framework, USAID's Product Life Cycle, and CGIAR's Scaling Readiness, based on six social theories related to AIS. This process is outlined below and illustrated in Fig. 1.

1. **Identification of applicable social theory**: The initial step involves identifying relevant social theories that, while not exhaustive, are pertinent to the AIS. This stage sets the theoretical foundation for the analysis by selecting theories that offer insights into equity and social dynamics within AIS.
2. **Selection of case studies**: Three case studies of AIS frameworks or AR4D tools currently in use are chosen. These case studies serve as practical examples to ground the theoretical analysis and provide a basis for examining how AIS frameworks or AR4D tools operationalize (or fail to operationalize) social considerations.
3. **Analysis through a social equity lens**: An analysis is conducted to uncover potential gaps in how the selected AIS frameworks or tools account for social dimensions, utilizing the identified social theories as an equity lens. This analytical process aims to reveal shortcomings or oversights in addressing social equity within the context of AIS.
4. **Thematic coding of identified gaps**: The gaps identified through the analysis are then categorized into themes. This step involves abstracting from specific instances to identify broader patterns or themes that capture the nature of the gaps in addressing social dimensions within AIS.
5. **Creation of EPs based on thematic responses**: EPs are formulated in response to identified themes. These principles are derived directly from the social theories applied in the analysis and an understanding of the tools and strategies available to AR4D actors. The EPs are designed to address the identified thematic gaps, providing a normative framework to enhance the social equity dimensions within AIS.

3.2. Input and data processing

3.2.1. Data collection

This study analyzes secondary data, with collection involving

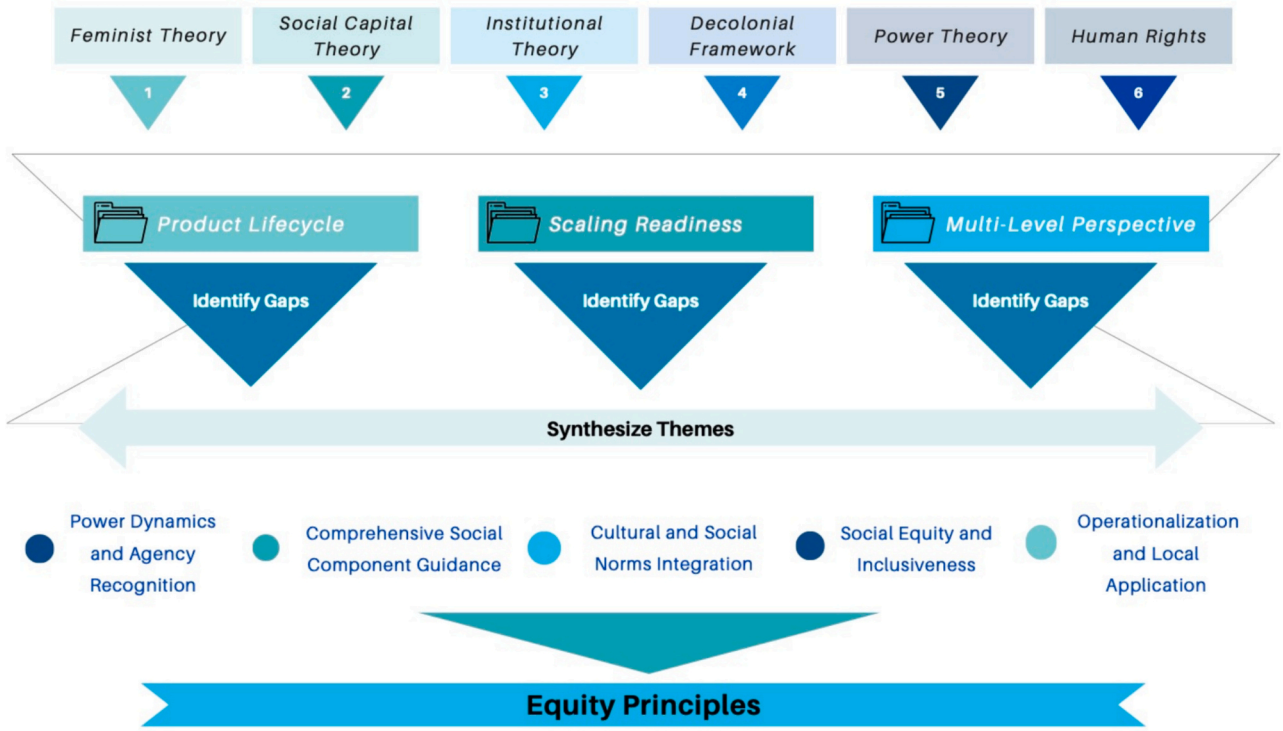

Fig. 1. Method to develop EPs. Figure created by Kristen Becker, University of California, Davis.

literature reviews on the characteristics of these three case studies (Fig. 2). We identified and examined relevant published and gray literature that provided information, critiques, and evidence related to the AIS theories and frameworks and AR4D tools to determine their alignment with social theory aimed at social transformation. The review included additional literature on social transformation and unintended social and environmental consequences due to AR4D innovation and scaling if deemed relevant to the research questions. The AIS document analysis facilitated a deeper understanding of the area under investigation while identifying and mapping relationships among major actors, their organizations, and the agencies involved in addressing these concerns. Finally, our literature review focused on social dimensions, the foundation for developing the EPs.

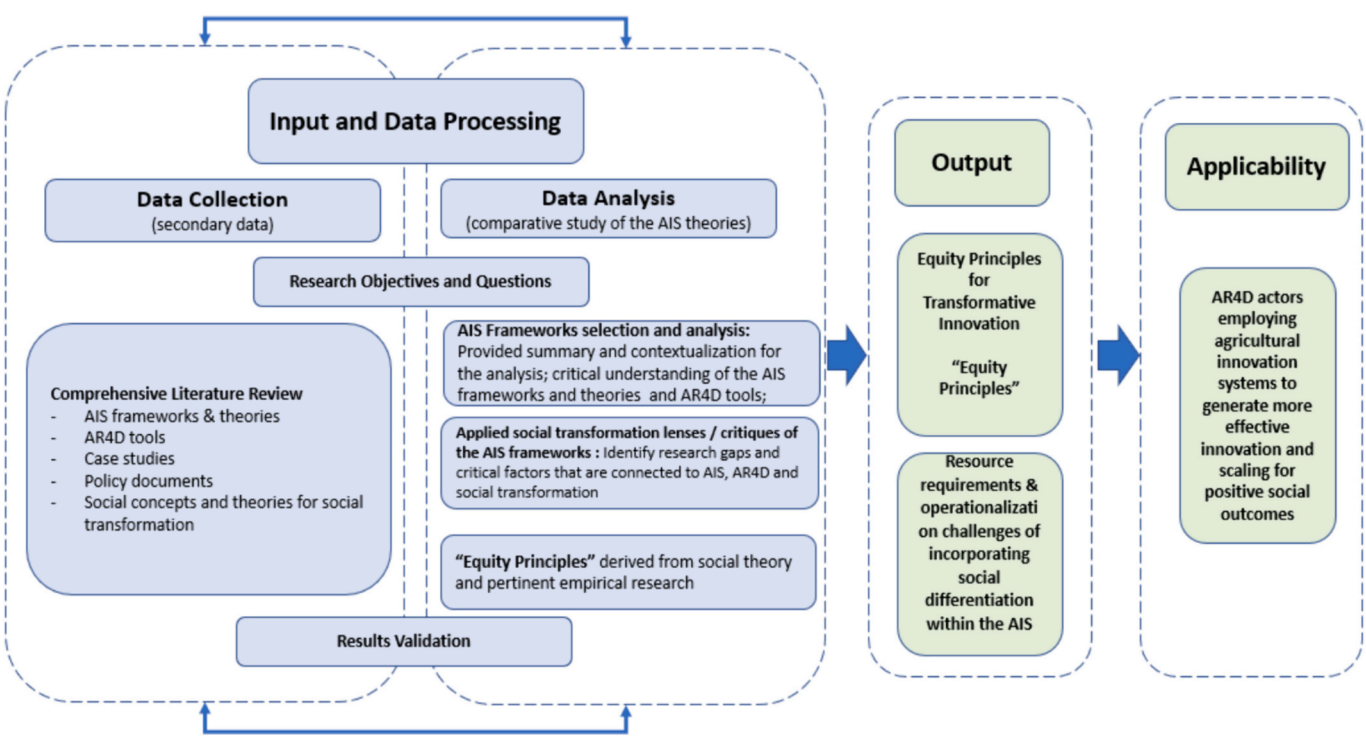

Fig. 2. Research methodology.

3.2.2. Data analysis

Combining multiple data sources can help facilitate a deeper understanding and overcome potential biases within qualitative research than when using a single method or data source (Cohen et al., 2000; Creswell and Miller, 2000; Koch, 1994; Shenton, 2004). We critically compared, integrated, and analyzed the literature review findings (both theoretical and policy literature), which were then cross-checked with information obtained from the various case studies (applied policy). This step produced a richer context, facilitated a comprehensive evaluation of evidence from scholarly sources, reduced bias, and built a coherent justification for the study recommendations. Where applicable the case study description and analysis were ground-truthed with the relevant creators and managers. The analysis results of each case study follow the structure below.

- A summary of the framework or tool is provided to contextualize the analysis.
- The strengths and opportunities of each case study are explored from a theoretical perspective.
- An equity analysis of each case study is conducted from a practical standpoint to achieve positive social outcomes. This includes an analysis to uncover potential gaps in how the selected AIS frameworks or AR4D tools account for social dimensions.
- The resource requirements and operationalization challenges of incorporating social differentiation within the case study is evaluated.

These full analyses are available in Appendix A and provide insights and recommendations for strengthening these case studies and their applications for positive social outcomes. However, for this paper's purposes, only the summary and equity analyses are below.

3.3. Case study selection and analysis

An AIS framework, the MLP Framework and two AR4D tools, USAID's Product Life Cycle tool, and CGIAR's Scaling Readiness tool, were chosen as case studies to explore any common omissions or gaps they may have in operationalizing social transformation. Selection was based on their frequent application in various projects, initiatives, and related research worldwide (Sengers et al., 2019). Each was analyzed for its ability to achieve effective social outcomes and progress toward the 2030 SDGs.

The reach of these frameworks and tools spans different levels and scales within AR4D projects. For example, the MLP is frequently used to assess agri-food innovation systems and model transitioning processes (El Bilali, 2019; Giagnocavo et al., 2022; Elsner et al., 2023; Leeuwis et al., 2021). They describe the comprehensive process from the innovation design (e.g., PLC) and innovation assessment (e.g., Scaling Readiness) to scaling pathways and transitioning processes (e.g., Scaling Readiness, MLP). They also open the opportunity to analyze these frameworks from the micro level (i.e., an innovation product) to the macro level (i.e., the structural landscape) and from downstream to upstream actors. Using the social theory lenses to analyze these three case studies, we illuminate the gaps in social transformation angles from both short- and long-term processes and cross-cutting perspectives. In our analysis of the MLP, PLC, and Scaling Readiness, we employed multiple social theory concepts, elaborated upon in the subsequent section. We aimed to use these social lenses to evaluate how effectively inclusivity and responsibility in innovation and scaling were encouraged in these case studies. We also sought to understand their potential impact on achieving positive social outcomes.

3.4. Applied social transformation

3.4.1. Selecting social theories

We selected the six social theories below for their relevance and applicability in explaining deep-rooted global power imbalances and local social exclusion within the AIS frameworks and AR4D tools' landscapes (Table 1). These theories offer critical insights into power dynamics, social equity, and structural inequalities, all essential to ensuring that innovation processes benefit marginalized populations in low- and middle-income countries. For example, Feminist, Decolonial, and Power theories provide frameworks to understand and challenge patriarchal structures, colonial legacies, and power imbalances that perpetuate marginalization globally. Institutional Theory illuminates the role of institutional norms and practices in shaping social behavior, while Human Rights Theory emphasizes the rights and dignity of all individuals within innovation processes. This selection reflects the diversity of the authors' backgrounds and experiences, ensuring an appropriate application of social theory to the case studies. Other social theories, such as Critical Race Theory, Conflict Theory, or Structural Functionalism, could be applied, and we would encourage similar analyses using these perspectives. The theories included provide a foundation for understanding the potential contributions of social theory to AIS. Further exploration may reveal additional insights or perspectives that enhance our understanding of responsible innovation and scaling processes, and the resulting outcomes.

3.4.2. Application

As a response to deep-rooted and widespread social exclusion, scholars have delved into the social norms and systems humans interact with to formulate social theories. These social theories and concepts criticize structural inclusivity challenges that are common worldwide. They offer insight into how AIS frameworks and AR4D tools can be more

Table 1
Social theories included in the analysis and their application in AR4D.

Social Theory	Core Concepts	AR4D Illustrative Cases In Practice
Feminism	Gender equality Social justice Intersectionality	Incorporating gender analysis into research design, data collection, and analysis; promoting gender-sensitive practices and technologies; reflecting on team diversity.
Social capital	Social relations Resources	Conducting network assessments and designing interventions that strengthen local ties and structures; promoting spaces for social interaction and community-based approaches.
Institutional	Social structures, norms, and cultural influence on behavior	Promoting equitable access to resources and decision-making; aligning agricultural services with local norms and values for effective community empowerment; structural frameworks for interdisciplinary research approaches.
Decolonial	Colonialism Indigenous rights Social justice	Respecting First Nation's sovereignty, knowledge, and practices; embracing exit strategies to ensure self-determination in the design and implementation process, increased local research leadership.
Power	Power dynamics Power distribution Governance	Engaging with marginalized groups during research design and implementation; addressing power imbalances in agriculture value chains; promoting equitable access to resources and markets for all actors, including women and small-scale farmers.
Human rights	Rights	Ensuring that agricultural practices and interventions do not infringe upon the rights of local communities, especially the vulnerable and marginalized.

socially inclusive with more equitable outcomes. We apply these social theories to the identified case studies to determine whether there are gaps in AR4D actors' understanding of relationships and leverage points within upstream or downstream innovation and scaling processes.

Feminist Theory initially advocated for gender equality but now have been extended to challenge all systems of oppression, including those based on race, class, sexual orientation, and other identities. Feminism challenges patriarchal structures and norms that limit socially marginalized populations' opportunities and contributions to society (Hooks, 2015). Intersectionality, a core feminist concept, acknowledges that social categorizations such as race, class, and gender are interconnected, creating overlapping and interdependent systems of discrimination or disadvantage (Crenshaw, 2005). Feminist perspectives advocate for the inclusive participation of women and marginalized groups in decision-making processes and the recognition and appreciation of their knowledge and roles.

Social Capital Theory suggests that social networks have value and that social relations represent a form of capital that can contribute to the success or failure of individuals and groups. These networks include relationships of trust, mutual understanding, shared values, and behaviors, all enabling cooperation within or among groups. These mutual understandings can confer significant power and influence on certain individuals or groups within a community. Conversely, those with less social capital may be marginalized or excluded from these processes (Coleman, 1988).

Institutional Theory explores how structures, routines, norms, and cultures within organizations or wider social systems shape social behavior. Central to this perspective is the concept of institutional power—the capacity to influence, direct, or control both the behavior of others and the course of events within a social system. Institutional theory also underscores how cultural norms contribute to creating and maintaining institutions. Culture involves shared beliefs, values, and norms that members of a society use to interpret experiences and generate behavior. These cultural elements are often deeply embedded in social systems and are transmitted across generations (Scott, 2013).

Decolonization frameworks respond to the historical legacy of colonialism and its impact on power structures, knowledge systems, and practices (Itchuaqiyaq and Matheson, 2021; Trisos et al., 2021). Scholars emphasize that the core of decolonial work is to support, respect, and restore the sovereignty of Indigenous peoples, lands, and knowledge (Itchuaqiyaq and Matheson, 2021). Within AIS, decoloniality highlights the need to challenge the dominance of Global North scientific knowledge and practices and recognize and value diverse knowledge and expertise (Leach et al., 2008). It also prioritizes and empowers local communities collectively, rather than centering around individual leaders. Decolonial frameworks urge decentralized power and promote local and regional control (Ramos-Cortez and MacNeill, 2023).

Power Theory examines how individuals and institutions exert authority and control over resources and decision-making processes. Power itself is a "relationship" between agents (Dahl, 1957) characterized by: a) the source of power, b) the means to exercise the power, c) the amount of power, and d) the scope of the power. A critical perspective on power can help identify and address power imbalances that limit the participation and contributions of these groups (Wigboldus et al., 2016). Power Theory also speaks to who has the agency to make decisions and receive benefits from them within AIS frameworks. It has even been acknowledged that processes of inclusion "inevitably force considerations of power" (Stilgoe et al., 2013 p. 1571). For this reason, recent research has also stressed the notion of empowerment as a way to understand how to gain a "sense of power." Empowerment is then defined as "the process of gaining capacity along three dimensions: a) access to resources and institutions, b) strategies to mobilize them, and c) willingness to do so" (Avelino, 2017 p.512). Within AIS, power dynamics can influence the distribution of resources, opportunities, and benefits among different actors, including marginalized groups.

Human rights recognize the societal agreements about individual and societal human needs for all, locally and globally. They can provide a normative perspective on innovations, prioritizing those that can respond to contextual social requirements. They also propose that both innovators and beneficiaries have duties and rights. This new classification of innovators as public or private has led to new activities from private innovator institutions and their beneficiaries (Land and Aronson, 2018, 2020).

3.4.3. Result validation

The EPs are formulated in response to the gaps identified in the case studies. This response, which is grounded in a practical ethos that can be applied widely, draws upon three major sources of knowledge:

1. **Social theory:** Each selected theory illuminates the interaction dynamics among various social groups, providing insights into addressing structural challenges within AIS, specifically within downstream and upstream processes.
2. **Author expertise and available tools:** The researchers represent various disciplines, come from varying geographies (both the Global South and North), and possess a range of professional and applied research expertise related to responsible research, innovation, scaling, and AIS. This helped facilitate a non-exhaustive but comprehensive analysis of the current tools available, language that resonates with other AR4D actors, and a foundation rooted in established best practices and knowledge.
3. **Personal observation**: Drawing from our unique perspectives as women from diverse intersectional backgrounds in AR4D, we have noted firsthand how upstream processes often mirror the disparities highlighted in our equity analysis, such as overlooked power dynamics. We suggest actionable strategies within the EPs, such as fostering horizontal partnerships to respond to these issues through a unified approach.

This diverse foundation ensures that our analysis is comprehensive and sensitive to the nuanced needs of AR4D actors operating across varied contexts, serving as an initial framework for further investigation and potential adoption within the AR4D community.

A "peer debriefing" process was undertaken to ensure the integrity and effectiveness of the proposed EPs (Spall, 1998). This process involved informally presenting the principles to fellow scholars well-versed in the research topic (Spall, 1998). For instance, early versions of the EPs were shared with AR4D colleagues for individual feedback, and several workshops on the EPs were held at conferences or workshops where relevant stakeholders were in attendance, such as the 2023 CGIAR Gender Conference and 2023 USAID Feed the Future Innovation Lab Directors meeting. This dialogue fostered critical critique, in-depth discussions, and valuable feedback that was used to refine the principles. As highlighted by Jabareen (2009, p. 55), "a theoretical framework representing a multidisciplinary phenomenon will always be dynamic and may be revised according to new insights, comments, literature, and so on." Although the extant literature on this specific topic is limited, the applied research methodology of data collection and analysis provides a comprehensive understanding of the subject matter. This methodological approach enhances understanding and offers actionable recommendations for actors within AIS and AR4D contexts.

4. Results

4.1. Question 1: Do AIS frameworks and AR4D tools give sufficient attention to social theory, and if not, how can social theory methods address frequent oversights?

Below, we summarize the results of an equity analysis of the MLP, PLC, and Scaling Readiness. For more detailed results, see the appendix. We consider the MLP model more theoretical and considered often as an AIS framework, while Scaling Readiness and the PLC are AR4D tools. We

want to acknowledge that it is nearly impossible to practically operationalize an entire AIS framework into an AR4D tool. Thus, AR4D tools, by necessity, omit certain systems' complexity to achieve practical goals. We do not aim to comment on the effectiveness of these approaches against one another; we assume they all advance AR4D thinking and contribute to positive development goals. This analysis aims to highlight where more social theory can be included to increase each framework or tool's effectiveness for positive social impact.

4.2. MLP framework

Summary: The MLP is a comprehensive framework that encompasses various socio-technical aspects of the system and is suitable for application in the agri-food sector. This socio-technical system approach considers three elements as part of the system: landscapes (macro-level characteristics of a system), regimes (structures, institutions of a time and place), and niches (the individuals developing new ideas) (Geels, 2011). According to this framework, a system transformation change results from the interactions and alignment of multiple transition pathways through time (Geels and Schot, 2007).

Equity analysis: The MLP is based on a socio-technical systems approach that blends technological, sociological, and institutional theories, but it has been criticized for its treatment of agency (Geels, 2011; Geels and Schot, 2007). The MLP underpins the role of agency in transitions but does not address power and political relationships (Genus and Coles, 2008; Smith et al., 2005). Furthermore, applying these conceptual levels empirically poses a challenge in terms of clarity, making it difficult to apply possible insights that result in action in practice (Smith et al., 2005). Geels has acknowledged that the MLP does not yet differentiate between global and local agency levels, nor does it describe some agency types related to power, civil society, and cultural dimensions (Geels, 2011). The MLP could address these critiques by providing a more defined system scope and a better description of the agents involved in a transition from a power and/or cultural relationship perspective.

Entry points for social differentiation: An improved social differentiation analysis could benefit the MLP in two ways. Firstly, the scope of systems needs to be refined from a global to a local scale. This will provide actionable insights for stakeholders to drive sustainability transitions (Hölscher et al., 2023). This involves translating the dimensions of the MLP into a geographically based system and identifying specific agents at each level. Secondly, the MLP should incorporate an analysis of agents or stakeholders, focusing on their power dynamics and cultural backgrounds. Understanding the historical and cultural roles of agents rather than just mapping them geographically, is crucial to understanding their power. Understanding the agents provides insight into their behaviors, intentions, interrelations, agendas, interests, and influences (Brugha and Varvasovszky, 2000), shedding light on the dynamics of this process and what is feasible from an agent perspective.

4.3. Emerging gaps

- **Power dynamics and actor agency**: The MLP Framework does not sufficiently address how power and political relationships influence transition processes. Additionally, it lacks agency related to global and local, civil society, and cultural dimensions adequately.
- **Contextual operationalization**: The MLP faces challenges in providing clear operational guidelines adaptable to various socio-cultural and geographical contexts.
- **Data availability and quality**: High-quality, context-specific data is crucial for identifying promising niches and understanding the intricacies of existing regimes.
- **Expertise and interdisciplinary collaboration**: The MLP's emphasis on socio-technical transitions necessitates a broad range of expertise spanning technical, social, and environmental domains to navigate the complexities of changing systems.

Gaps derived from the following social theories: Power Theory; Feminist Theory, Human Rights Theory, and Social Network Theory.

4.4. Product life cycle tool

Summary: The PLC approach focuses on the different stages of an agricultural product's life, from research and development to its introduction, growth, maturity, and eventual decline. The Product Life Cycle (PLC) represents the latest enhancement to USAID's suite of scaling tools, augmenting the Agriculture Scalability Assessment Tool (Kohl and Foy, 2018) and serving as a critical component of a decision-making tree designed to aid innovation managers with scaling decisions (Kohl and Foy, 2018, p. 49). This approach emphasizes the need for continuous innovation and adaptation to changing landscape conditions, consumer preferences, and environmental factors to remain competitive and sustainable. The PLC aims to facilitate innovation adoption by ensuring a good fit between product and market and coordination between various actors.

Equity analysis: The PLC process subtly engages various actors and networks yet lacks transparency in delineating ownership and leadership, obscuring the diversity among decision-makers. The PLC's approach, primarily designed for technological innovations, struggles to adapt seamlessly to social and institutional innovations, despite their inclusion in current practices. Many innovations necessitate complementary innovations or "innovation packages" for effective use, tailored to different user groups. Although the PLC acknowledges the role of complementary innovations, it does not place a strong emphasis on their evaluation. Moreover, the PLC's oversight of alternative scaling approaches, which might better serve marginalized groups through public sector and grassroots initiatives, underscores a potential bias toward methodologies and strategies commonly adopted in high-income marketplaces (although USAID broader guidance gives insight into other scaling pathways: Kohl and Foy, 2018, p. 49). Such a bias threatens to diminish the autonomy, expertise, and traditional knowledge of local and Indigenous communities, thereby reinforcing longstanding power disparities. Without a concerted effort to rectify these imbalances, there's a risk that dominant groups will consolidate control over benefits and decision-making processes, exacerbating the exclusion of less empowered stakeholders. Further, in its current version there is no explicit or required end-user base assessment. It is important to note that the PLC is a new tool, and USAID has intentions of incorporating aspects of "*Scaling for Widespread Adoption of Improved Technologies and Practices*" which speaks to actor agency, public-private scaling pathways, gender inclusion, and more end-user landscape assessments (USAID, 2023).

Entry points for social differentiation: The PLC stages provide opportunities to involve diverse stakeholders, to ensure marginalized voices are considered throughout. As such, this framework can help identify stakeholder involvement at each development stage, revealing actors' existing or missing capabilities at each step. Adaptations to the PLC could facilitate analysis of other more organizational or social innovations. Finally, monitoring reflective assessments before, during, and after the project could encourage trade-off reflection and reactions. Additionally, introducing a mechanism to gauge how scaling affects different user groups could enhance the PLC's social impact.

Emerging gaps:

- **Power dynamics:** lacks comprehensive guidance on addressing power dynamics in upstream research processes or downstream complex contextual challenges.
- **Innovation relevance and contextualization:** In the early stages of innovation development (e.g., the "research and development" and "introduction" stages). The PLC's emphasis on private-sector mechanisms might neglect the cultural and institutional norms that shape social behavior, limiting its effectiveness in incorporating local realities.

- **Inclusiveness and responsiveness:** The discretionary inclusion of tools for analyzing power dynamics and social inclusion may lead to ineffective strategies for overcoming barriers for certain groups.
- **Incomplete guidance on considering social components:** Stage-gates do not incorporate social components or reflections on unintended consequences.
- **Social inequity reinforcement:** The framework may reinforce social inequities by privileging innovations developed within certain paradigms, potentially sidelining marginalized populations.

Gaps derived from the following social theories: Institutional Theory, Power Theory, Feminist Theory, Decolonial Theory.

4.5. Scaling readiness tool

Summary: The CGIAR's Scaling Readiness approach is a framework that aims to assist AR4D (organizations, programs, and projects) interventions in developing, implementing, and monitoring the scaling of innovations in a structured and evidence-based way. It assesses an innovation's readiness or maturity for scaling and the actions that could improve or accelerate the process. Core concepts include limiting factors to scaling, innovations scale with other innovations (e.g., "core" and "complementary"), and that successful scaling is a function of innovation readiness and innovation use (Sartas et al., 2020). Therefore, the unit of analysis for assessing scaling readiness is the innovation package as a whole. The following steps comprise the Scaling Readiness framework:

- Step 0: Fit-for-purpose. Determine that the use of the tool is benefiting the research project with adequate resources and time for implementation.
- Step 1: Characterize. Understand the innovation, core and complementary components, context, and stakeholders involved.
- Step 2: Innovation Profile Development. Diagnose and evaluate the scaling readiness and current use of each innovation in the package.
- Step 3: Innovation Packages. Strategize and analyze the strategic options to increase the scaling readiness with stakeholder analysis for potential solutions and partnerships.
- Step 4: Agree. Seek agreement among stakeholders on scaling investment decisions and facilitate learning and negotiation.
- Step 5: Scaling Strategy. Identify emerging windows of opportunity, tipping points, and the process of monitoring, evaluation, and learning.

This framework examines factors such as the innovation's effectiveness, feasibility, acceptability, and potential for impact at different scales. The Scaling Readiness framework was developed in 2018 as a conceptual framework (Sartas et al., 2020) and decision-support process (Schut et al., 2022) and has since been in continuous development as a tool that AR4D actors can use to understand how to scale their innovations. This is usually done over a one- to three-day workshop with a trained facilitator. Currently, Scaling Readiness is being transformed to be used throughout the CGIAR as an innovation management and curation database, referred to as the Innovation Packages and Scaling Readiness (IPSR). This analysis focuses on the Scaling Readiness concepts.

Equity analysis: Scaling Readiness includes tools specifically designed to analyze and facilitate the scaling of innovations. Thus, some of the strengths of Scaling Readiness from a technological perspective are also part of its critiques from a responsible innovation perspective. As the framework focuses on the innovation itself, it does not yet offer comprehensive guidance to reflect and decide upon certain social components during the innovation's scaling development phases. These include understanding power dynamics, considering complex contextual challenges when developing an innovation, and reflecting on unintended consequences and outcomes of the scaling of the innovation. In addition, it does not explicitly address gender and other vulnerable groups and minorities' needs as part of the selection and monitoring of the evaluation. This last point provided the impetus for creating the GenderUp Tool, which was developed to be used independently or jointly with Scaling Readiness to foster gender-responsible scaling (McGuire et al., 2023).

Complementary analytical tools like social network analysis, power mapping, and intersectional gender assessment can be used to understand power dynamics during stakeholder identification and engagement phases, e.g., Steps 1, 3, and 4. However, their application is optional at the discretion of the scaling team. Scaling Readiness, as stated by the authors in Sartas et al. (2020), by design does not gauge the inclusivity and responsibility of an innovation's scaling. However, by including this tool as part of a larger initiative, the CGIAR's Innovation Platform for Scaling Readiness, efforts have been made to analyze alternatives to systematically enhance gender and diverse stakeholder engagement. This includes a manual for trainers who will use Scaling Readiness that recommends using participatory tools, along with questions to prompt gender reflection. Although Scaling Readiness provides social inclusion entry points, it grants upstream actors the choice to explore or ignore authority structures and biases, potentially limiting the effectiveness of strategies to address scaling barriers like unequal resource access. Not making these tools integral to the process downplays their importance in driving social transformation.

Entry points for social differentiation: Beyond understanding the technological process when scaling an innovation, other elements that assess the readiness of an innovation in a context could be incorporated to consider social differentiation. Addressing power dynamics in the agricultural sector is a complex challenge. Nonetheless, enhancing the framework's focus on discerning and analyzing power dynamics (considering both top-down and bottom-up approaches), acknowledging trade-offs, developing safeguards, and recognizing intersectional vulnerabilities would be critical to increasing this tool's impact for effective social transformation. Additionally, the tool could offer broader prompts emphasizing market-oriented innovations and recognizing local knowledge and practices with scaling potential. By focusing on the target landscape and its interactions with innovation processes, Scaling Readiness can better promote the integration of social theory for effective, contextual scaling.

Incorporating assessments of social differentiation into the evaluation of an innovation's readiness for scaling can enrich the understanding beyond just the technological aspects. Tackling power dynamics within the agricultural sector is notably intricate. An enhanced focus on dissecting and addressing power dynamics through both top-down and bottom-up perspectives, acknowledging necessary trade-offs, instituting safeguards, and recognizing intersectional vulnerabilities, is essential for amplifying the tool's capacity for driving meaningful social change. Furthermore, the tool could introduce broader prompts that not only highlight market-driven innovations but also acknowledge and valorize local knowledge and practices with potential for scaling. By concentrating on the interplay between the target landscape and innovation processes, Scaling Readiness could more effectively leverage social theoretical frameworks to ensure contextually apt and impactful scaling strategies.

The different steps within the Scaling Readiness framework present various opportunities to consider social differentiation. Step 1 of the analysis, when characterizing the stakeholders, their networks, and their interventions, presents a critical opportunity to understand the users of the innovation (including women, youth, and other vulnerable groups in societies) and their needs. This assessment could go beyond the understanding of the technical capacities to comprehend the current mindsets and capacities of the people that are targeted to adopt the innovation in the future and develop relationships and connections with them to understand their innovation concerns. If necessary, capacity-strengthening strategies could be included as part of the scaling process that Scaling Readiness is driving to bring more people in as part of the process. The

training could be for innovators, as is currently being developed, and other external agents that can be more neutral to the power dynamics.

Emerging gaps:

- **Power dynamics:** There is a lack of comprehensive guidance on addressing power dynamics in upstream research processes or complex downstream contextual challenges.
- **Innovation relevance and contextualization:** This is essential for the "Characterize" and "Innovation Profile Development" steps, where aligning the innovation with the context and needs of the stakeholders is crucial to ensure the effectiveness and acceptability of the scaling process.
- **Inclusiveness and responsiveness:** The discretionary inclusion of tools for analyzing power dynamics and social inclusion may lead to ineffective strategies for overcoming barriers for certain groups.
- **Incomplete guidance on considering social components:** The focus on innovation's technical maturity overlooks critical social components, such as power dynamics and the needs of vulnerable groups during the scaling process or reflection on unintended consequences.
- **Social inequity reinforcement:** The framework may reinforce social inequities by privileging innovations developed within certain paradigms, potentially sidelining marginalized populations.
- *Gaps derived from the following social theories: Institutional Theory, Power Theory, Feminist Theory, Decolonial Theory, Decentralization Theory, and Social Capital Theory.*

4.6. Cross-cutting equity analyses gap themes

Below we consider the emerging gaps of the above social analyses and group them into themes.

(i) **Power Dynamics and Agency Recognition:** Gaps relate to the insufficient addressing of power dynamics, the lack of detailed description of agency types, and the framework's failure to adequately consider power relations and political structures. AIS frameworks and AR4D tools often overlook the positionality of primary drivers. While the MLP emphasizes niche ideas influencing regimes, there is limited introspection on the relationships between architects of change and their relationships with the end-users. Agency tends to remain with the upstream decision-makers, overlooking other actors and the effects of the innovation on them. Without overtly acknowledging dominant players, these frameworks risk sidelining marginalized stakeholders (Dobbin et al., 2023). *This theme captures the need for AIS frameworks and AR4D tools to explicitly incorporate analyses of power and agency to ensure equitable transitions and innovation processes.*

(ii) **Comprehensive Social Component Guidance**: Gaps relate to providing comprehensive guidance on addressing critical social components, such as understanding complex contextual challenges, reflecting on unintended consequences, and considering the needs of vulnerable groups during the innovation and scaling process. Mechanisms to support innovation and interventions that could produce more sustainable systems capacity, such as education and institutional capacity, are not acknowledged. *This theme calls for AIS frameworks and AR4D tools to encompass a broader view of social components, ensuring a holistic approach to innovation scaling that addresses potential social and environmental impacts.*

(iii) **Cultural and Social Norms Integration:** Gaps highlight the oversight of cultural and institutional norms and the frameworks' limitations in capturing the complexities of grassroots realities. To achieve positive social outcomes and ultimately reach social transformation, understanding the intricate web of local interactions, from community group dynamics to regional politics, is critical. Current frameworks might not delve deep enough, potentially bypassing local context or user-driven innovation. This can result in innovations that may be technically sound but misaligned from a socio-cultural or infrastructure perspective. *This theme underscores the importance of integrating cultural and social norms into AIS frameworks and AR4D tools to ensure they respond to and reflect the diverse realities of all stakeholders.*

(iv) **Social Equity and Inclusiveness:** Gaps related to the exacerbation of social inequities and the frameworks' potential to privilege certain innovations over others, sidelining marginalized populations. Additionally, gaps related to the case studies' flexibility in addressing social inclusion lead to ineffective strategies for overcoming barriers for specific groups. The distribution of benefits and burdens among communities and the environment is not evaluated. None of the case studies consider a stratified perspective of benefits or burdens of niche ideas or scaling of innovation. This approach overlooks the perspectives of marginalized communities, as the trade-offs might be felt most acutely by these groups, potentially deepening inequalities or perpetuating societal hierarchies. For instance, focusing on market-driven innovation can jeopardize AR4D actors' intent on systems-level change. A slight shift to 'inclusive' market solutions with a broader perspective that incorporates community-centric or organizational innovations could ensure that interventions are socially equitable and contextually relevant. *This theme stresses the imperative for AIS frameworks and AR4D tools to proactively promote social equity and inclusiveness, prioritizing the voices and needs of marginalized and vulnerable groups.*

(v) **Operationalization and Local Application:** Gaps concerning the lack of clear operational guidelines for applying frameworks and tools in diverse contexts, including the challenge of translating conceptual pilots into sustained local action. *This theme emphasizes the need for AIS frameworks and AR4D tools to provide clear, actionable guidance that can be effectively implemented across different socio-cultural and geographical settings; and highlights the need for regional expertise and leadership.*

These emerging themes within our case studies allow us to consider what practical social considerations we might consistently apply to other AIS frameworks and AR4D tools supporting the SDGs' achievement.

4.7. Equity principles for social transformation

Question 2: Given common emerging themes, can we establish standard principles that AR4D actors can adopt within the AIS frameworks or AR4D tools they use?

The following EPs respond to recurring gaps observed within each case study's equity analysis. Considering the common structures of AIS frameworks and AR4D tools, these EPs aim to identify key leverage points to generate more effective innovation and scaling for positive social outcomes through an inclusive systems approach. Focusing on upstream and downstream AIS processes, we categorize the EPs into seven topics and provide mechanisms to employ them. The principles are the foundation for shaping and achieving the desired outcomes of AIS in alignment with the SDGs but could be translated and used in other sectors as well. They allow for flexible application without a strict sequence.

Equity Principles

I. **Recognize AR4D Power Dynamics**

AR4D actors experience different power dynamics in time and space that shape how they interact and the outcomes of the innovation process. Recognizing who the stakeholders are with and without power, the amount of power they have, and how the power is exercised will clarify the directionalities of the selected and

designed innovations. AR4D actors, especially within upstream functions, should acknowledge and manage the influence, biases, and composition of innovation and scaling teams to balance—or at least acknowledge—control-oriented approaches or leverage empowering ones.

Mechanisms to achieve this:

- **Map stakeholder power relations**: Identify and acknowledge stakeholders with and without power. Understand the source, amount, and means of exercising their power. Ensure project leaders understand the social dynamics of different stakeholders contributing to the innovation process.
- **Analyze opportunities to balance power:** Evaluate feasible and desired pathways to balance power in the innovation process to achieve the SDGs agri-food system goals.
- **Manage conflict reduction:** Jointly develop a set of minimum rules for addressing power changes in stakeholder meetings, communications, and working together. Examples of rules are respect, listening to every individual's voice, and active listening. Sensitize stakeholders with common biases and support the team to use techniques to identify, reflect on, and mitigate such biases.
- **Ensure appropriate authorship:** Encouraging ideas from all actors is valuable, equally as valuable is ensuring all ideation gets appropriate recognition. Develop an equitable system for knowledge acknowledgement. See frameworks such as the Civic Laboratory for Environmental Action Research (Civic Laboratory, 2016).

Responding to theme: Power Dynamics and Agency Recognition

II. **Define Goals, Anti-Goals, and for Whom:** AR4D actors must set project objectives that articulate the intended changes in the external environment and identify the specific groups targeted by these changes. It is also crucial to define and communicate the project's anti-goals, which are the undesirable outcomes the project aims to avoid. These anti-goals and necessary actions can be determined through ex-ante processes that account for context-specific considerations and relevant on-the-ground experiences. Given the inevitability of unintended consequences, AR4D actors, with their frequent goals of elevating vulnerable populations, can prioritize fundamental anti-goals such as "avoid causing harm to marginalized groups" or "prevent exacerbating community inequities."

Given the possibility of unforeseen negative outcomes, establishing comprehensive monitoring and evaluation systems is essential for detecting and addressing unintended negative consequences. This approach enables AR4D actors to proactively manage and mitigate some potential or likely adverse impacts rather than passively experiencing them, thereby exerting control over what they aim to avoid and implementing strategies to prevent and respond to such scenarios effectively.

Mechanisms to achieve this:

- **Appreciative Inquiry and theory of change:** Challenge identification and possible solutions sets should be co-created through appreciative inquiry approaches. Analyze the desired and undesired impact, outcomes, and outputs of the innovation using scenario methodologies that examine both scenarios.
- **Anticipate intended and unintended outcomes:** Anticipating intended and unintended outcomes, both qualitatively and quantitatively, before the scaling process can help actors adjust interventions. This will allow for minimal negative outcomes and ensure equitable distribution of social, economic, and environmental outcomes for different social groups.
- Include **exit strategies** to ensure communities self-determination in the design and implementation process. By prioritizing community autonomy, communities could challenge power imbalances by asserting their autonomy and reclaiming ownership over research initiatives.

Responding to themes: Operationalization and Local Application, Social Equity and Inclusiveness.

III. **Build Global "Horizontal" Partnerships**: Promote genuine collaboration toward common goals, ensuring transparent communication and representation across sectors. Horizontal partnership includes equal decision-making power and influence, mutual respect, shared responsibility, and appropriate credit for knowledge production, with all parties involved having an equal say and stake in the outcomes.

Mechanisms to achieve this:

- **Prioritize authentic representation**: Promote the authentic participation of underrepresented groups, such as women and minorities, in agricultural innovation, avoiding superficial inclusion. Address their specific needs in design, implementation, and monitoring. Ensure active engagement in decision-making and encourage autonomy in setting priorities and mechanisms.
- **Collaborate across sectors**: Foster partnerships and collaborations among various stakeholders, including ministries, governments, NGOs, the private sector, and local communities (e.g., through multi-stakeholder platforms). Encourage dialogue and knowledge-sharing to create comprehensive, contextually appropriate, and sustainable visions and strategies.
- **Encourage local leadership:** Prioritize local leadership, insights, and stakeholder involvement throughout the innovation process and connect with regional, national, and global agendas. Fund local institutions to lead research and provide expertise.
- **Respect and amplify local knowledge**: Engage local stakeholders early and consistently, leveraging their existing efforts, knowledge, and capacities throughout the innovation and scaling processes. Ensure that contributions are properly recognized, and knowledge and innovations are appropriately attributed.
- **Nurture spaces of interaction and collaboration:** Foster environments where stakeholders can engage in collective learning and action. Depending on the objectives and context, these can be communities of practice, on-site farm experiences, seed fairs, or living labs.

Responding to themes: Power Dynamics and Agency Recognition, Comprehensive Social Component Guidance, Operationalization, and Local Application

IV. **Acknowledge Social Differences among Innovation Users and Non-Users:** Users will have different abilities to use and benefit from an innovation. One innovation will benefit different groups of people differently. Different groups of people will need different resources, scale pathways, and complementary innovations to realize the positive effects of innovation. It is also important to understand the implications of innovation on non-users. For example, developing more sophisticated innovations can replace less sophisticated ones that marginalized populations depend on for income.

Mechanisms to achieve this:

- **Ensure team diversity and representation:** Ensure the diversity of the programming team in terms of different factors such as expertise (e.g., engineering, economics, sociology, nutrition, innovation), gender, and nationality, especially considering those from low- and middle-income countries.
- **Identify what innovation or system will be replaced**: What current system or technology will need to be downscaled or usage reduced to allow for an innovation to be used? What system and livelihoods are attached to it?

- **Understand user capital and resources:** Engage with local experts, community members, and stakeholders to gather insights and perspectives on user capital, including social, physical, natural, and financial capital.
- **Conduct context analysis:** Assess the drivers and barriers shaping the past and present cultural, social, economic, and environmental systems. Find entry points using social lenses as Illustrated in the methodology.
- **Consider how innovation will change once introduced.** Innovation and its benefits will develop dynamically in response to power dynamics in a community. An innovation that is intended to, and perhaps does, benefit one group of people when introduced may not continue to do so over time, as community members with more power due to resources, networks, etc., usurp control.

Responding to themes: Power Dynamics and Agency Recognition, Cultural and Social Norms Integration, Social Equity and Inclusiveness

V. **Innovate and Curate Innovation Appropriately:** Innovate and scale innovations that disproportionately benefit marginalized populations, including at-risk international social groups, and respond to local priorities. This includes proper consideration at stage-gating and data collected for innovation stored within broad-scale innovation repositories.

Mechanisms to achieve this:

- **Elevate the underserved:** Include and prioritize innovation that have the potential to benefit low-resourced households, particularly considering access to inputs, education, and markets. This may mean that AR4D actors choose innovation and scaling pathways with a continuum of public/private mechanisms.
- **Innovate, stage-gate, and curate innovation for social inclusion:** Implement a review process at each innovation development phase to ensure projects specifically advance the interests of marginalized and at-risk groups. This approach requires detailed data collection and analysis on social impacts, ensuring innovations are curated with a commitment to social inclusion, addressing local priorities, and fostering equitable access for marginalized communities.
- **Consider innovations that optimize co-benefits:** Prioritize innovations that can bring multiple benefits, such as poverty reduction, ecosystem restoration, and gender equality.

  Understand the trade-offs: Innovation processes drive change, which will cause positive and negative impacts for different communities. Multiple innovation choices often coexist; opting for one often means sacrificing the benefits of another and creating winners and losers. Systemize the understanding of these trade-offs.

Responding to themes: Social Equity and Inclusiveness, Comprehensive Social Component Guidance

VI. **Assess Impact and Reflect**: Encourage ongoing monitoring, evaluation, and adaptation, emphasizing flexibility, long-term perspectives, and genuine representation of results and learnings.

Mechanisms to achieve this:

- **Do not "win-wash" results:** Do not only consider the overall benefits of innovation and scaling. Disaggregate data to get insights into the "winners" and "losers" and prioritize the options that can generate a more equitable distribution of social, economic, and environmental effects.
- **Consider flexible SMART goal strategy and update goals:** Consider SMART (Specific, Measurable, Achievable, Relevant, Time-bound) strategies to set goals and co-create them with upstream and downstream actors and intended innovation users when possible. However, goals should be considered as one assumption and might require revisiting for continuous reflection and adjustment. Systems change is more difficult to measure and often takes longer. Refer to updated quantitative and qualitative evidence and discuss with end-users, local stakeholders, and experts to update goals and anti-goals.
- **Implement holistic monitoring, evaluation, and learning:** Use a holistic approach in monitoring and evaluation, considering desired changes, causal impact pathways, and world views as foundational assumptions for continuous learning and reflection. Reflect on the project during times of both success and failure. Revisit the changes and process during innovation scaling to maintain the fundamental aim and inclusivity.
- **Employ a flexible approach in programming:** AR4D actors should acknowledge the complexity, uncertainty, and risk involved during the innovation scaling process. Therefore, it is necessary to employ a probabilistic approach in programming rather than fixed indicators and linear impact pathways.

Responding to themes: Comprehensive Social Component Guidance, Cultural and Social Norms Integration, Operationalization and Local Application

VII. **Develop Inclusive Systems Capacity:** Enhance AIS' structural and operational capacity to promote equity and inclusiveness at all levels of the innovation process, through increased capacity strengthening *of upstream* as well as downstream actors. Build an environment that supports sustained social transformation, acknowledging the importance of acknowledged biases (sexist, ethnic, geographic etc.), power imbalances, systemic resilience, and adaptability to effectively meet the needs of diverse communities.

Mechanisms to achieve this:

- **Invest in capacity strengthening and education:** Enhance educational programs and initiatives that empower AR4D actors to fully consider social implications within AIS. Prioritize skill and knowledge development that meets local needs and encourages community-wide participation and benefit from innovation processes. This should encompass training for both upstream and downstream AR4D actors.
- **Strengthen institutions:** Strengthen institutional frameworks and governance structures to support inclusive innovation and equitable scaling practices. This includes promoting resources, policies, and practices that encourage diversity, equity, and inclusion within organizations and innovation platforms.
- **Systemize Locally Driven Solutions:** Focus on creating systems that match challenges with solutions that are locally driven and appreciated, emphasizing outcome-based objectives tailored to local needs and leadership, rather than purely innovation-based objectives.

  Responding to themes: Comprehensive Social Component Guidance, Power Dynamics, and Agency Recognition.

Using the EPs when creating, curating, and scaling innovation will help AR4D actors be more effective in their applied research efforts. Table 2 highlights individual SDGs, relevant social concepts, and how innovating and scaling without considering social differentiation can impact these goals.

5. Discussion and conclusions

AIS frameworks and AR4D tools have advanced from linear models to represent more diverse ecosystems from both a theoretical and practical standpoint. However, to understand the leverage points within these systems to generate social transformation and contribute to achieving the SDGs, a critical next stage of advancements concerning social differentiation is necessary. Our results demonstrate how the current representations of some AIS frameworks and AR4D tools strip actors of their inherent power and agency relative to one another. This is a significant disadvantage as many of the challenges AR4D actors aim to

Table 2
Links between SDGs and social perspectives and the potential ramifications of overlooking these links.

SDG	Social Aspect	Specific Examples
SDG 1: No Poverty	Gender equity, feminization of poverty, and inclusivity	Ignoring these considerations could lead to innovations that disproportionately benefit wealthier individuals or groups, further widening income inequality and exacerbating poverty levels among vulnerable populations.
SDG 2: Zero Hunger	Addressing the needs of marginalized communities	Innovations that do not address the needs of marginalized communities, including women and racial and ethnic minorities, may not effectively contribute to food security and improved nutrition for these groups.
SDG 5: Gender Equality	Involving women and marginalized genders	Failure to involve women and other marginalized genders in the innovation process can perpetuate existing gender inequalities, limiting opportunities for their economic empowerment and restricting their access to resources, technology, and markets.
SDG 10: Reduced Inequalities	Decolonization, decentralization, Critical Race Theory perspectives	Ignoring these perspectives may contribute to reinforcing systemic inequalities between the Global North and South, as well as among racial and ethnic groups, hindering efforts to reduce inequality within and among countries.
SDG 12: Responsible Consumption and Production	Local and Indigenous knowledge	Agricultural innovations that do not incorporate local and Indigenous knowledge may promote unsustainable practices, jeopardizing long-term environmental health and exacerbating resource depletion. This could work against responsible consumption and production patterns.
SDG 13: Climate Action	Inclusivity, authentic representation	Innovation systems that do not prioritize inclusivity and authentic representation may fail to address the specific climate change adaptation and mitigation needs of vulnerable populations, who often bear the brunt of climate-related impacts.
SDG 15: Life on Land	Contextually appropriate solutions, collaboration with local communities	Failing to consider contextually appropriate solutions and collaborate with local communities could lead to innovations that have a negative impact on ecosystems and biodiversity, which could undermine efforts to protect, restore, and promote the sustainable use of terrestrial ecosystems.
SDG 16: Peace, Justice, and Strong Institutions	Inclusive decision-making processes, genuine participation	AIS can inadvertently contribute to social conflicts and tensions by not promoting inclusive decision-making processes and genuine participation. This is especially true when local communities feel disempowered or disenfranchised, hindering progress toward peaceful and inclusive societies.
SDG 17: Partnerships for the Goals	Cross-sector collaborations, open communication channels among stakeholders	Failure to establish cross-sector collaborations and open communication channels among stakeholders can impede the development of innovative, comprehensive, and sustainable solutions, limiting the potential for successful partnerships for achieving the SDGs.

address are social in nature. A growing and robust body of gender and social inclusion research (among other communities of practice, see CGIAR's Gender Platform) consistently demonstrates that failing to incorporate social considerations from both upstream and downstream perspectives can hinder AR4D actors' ability to create positive social change, and can contribute to anti-goals of further marginalization of already vulnerable populations. The analysis presented here shows how considering and operationalizing various social lenses in AIS frameworks and AR4D tools can better support innovation development and scaling efforts in achieving social transformation, thereby mitigating associated risks. In parallel, we propose a series of EPs for social transformation that offers a comprehensive framework to guide AR4D actors in creating more inclusive and impactful innovation systems.

While recognizing the logistical limitations of full consideration of all AIS aspects, our analysis of the case studies elucidates a common trend of focusing on functional components while overlooking power dynamics and agency within innovation processes. This trend aligns with EPs I and VII, which both highlight the significance of understanding decision-making dynamics, representation of marginalized groups, and the necessity of systems inclusivity. The analysis emphasizes the importance of going beyond technical innovation and actively involving women, ethnic minorities, and marginalized communities throughout the process. The analysis results highlight the discrepancies between current practices and the recommended approach for achieving equitable innovation outcomes.

The analysis revealed that innovation systems often center on scaling technologies without equal consideration for their diverse impacts on different user groups. This resonates with EPs II, IV and VI. The analysis illustrates how the selected AR4D tools prioritize scaling technologies based on their potential for widespread adoption. However, this approach can inadvertently sideline the equitable distribution of innovation benefits across user groups (Wigboldus and Brouwers, 2016). The prevailing focus on adoption rates and technical success often overlooks the broader social outcomes and impact disparities that might arise from scaling efforts. The equity analysis results underscore the imperative of shifting the focus from technological success to broader societal outcomes.

To enact significant transformation, it is necessary to examine and address the underlying structures of the system itself rather than merely focusing on inputs and outputs (McGuire et al., 2024; Meadows, 2008). Systems that have historically engendered inequities might not be inherently suited to devise effective remedies. Such systems could actually exacerbate the problems they aim to resolve by proposing insufficient or superficial solutions (Stroh, 2015). To disrupt the status quo, a critical approach is needed that challenges the existing norms and assumptions of the system and involves diverse actors and perspectives (see proposed challenges in Table 3).

In AR4D innovation and scaling activities, intricate power dynamics often pose challenges. Such dynamics may lead to donor preferences

focusing on mechanisms that may not contribute to positive social transformation. There may be a higher emphasis on natural sciences over social sciences, biases within AR4D teams based on gender, class, and location, and a lack of system-wide budgets to achieve positive social outcomes. Consequently, these biases can skew the identification of challenges, lead to unsuitable research designs and data collection, and ultimately negatively impact users and non-users during scaling activities. This inadvertent dominance of certain actors can diminish the efficacy of innovations by sidelining voices that may align more closely with ground-level realities, ultimately compromising the inclusiveness of the resulting innovations and scaling activity.

Additionally, the criteria AR4D actors use to define success directly shape the priorities of their AR4D tools in terms of activity and impact. While there's an ongoing emphasis on straightforward metrics like productivity and income boosts, there's insufficient discussion about the winners and losers as some innovations expand and others diminish. Even as literature evolves to encompass various scaling dimensions like "out," "up," "deep," and "in" scaling, the primary concern remains the extent of an innovation's use post-introduction (Carter et al., 2018; Moore et al., 2015; Sánchez Rodríguez et al., 2021). This narrow perspective on milestone achievement overlooks the distribution of benefits from such widespread use, e.g., income gains. It also overlooks the mechanisms by which these goals are achieved, prioritizing quick, potentially unsustainable, fixes. By zeroing in on sheer numbers rather than the overarching objectives of innovation, such as women's empowerment, scaling becomes an end, overshadowing the underlying social mission.

AR4D actors continually navigate the delicate balance between addressing immediate, tangible challenges and the broader, more intricate structural issues that underlie social disparities. The tendency to compartmentalize these challenges might offer short-term alleviation but often at the expense of holistic, long-term solutions. Effective innovation must resonate at institutional, cultural, and community levels. This might mean emphasizing education and leadership development, or inclusive financial capital schemes and market access for women, over, say, introducing a new yield-increasing seed variety. Community members must engage with an innovation in a dynamic, meaningful and integrated manner to upend deep-rooted social inequity. This type of innovation and scaling takes different sets of expertise and different timelines.

The challenges outlined are just some of the many obstacles AR4D actors face. This might explain why social theories and lenses are not fully incorporated into AIS frameworks and AR4D tools. This shortfall may lead to unintended outcomes and hinders progress toward social transformation objectives.

5.1. Challenges to operationalizing EPs within AR4D

Diverse levels of capacity mobilization: Implementing the EPs hinges on understanding the need for transformative capacity. Currently, AR4D actors are not fully equipped to innovate and scale with social differentiation or local adaptation in mind. The capacity and willingness to implement the EPs varies widely due to differences in context, resource availability, and previous experiences. Effective leadership and facilitation are crucial in multi-actor processes, yet there is frequently a shortage of individuals who have a transformative mindset and understand user perspectives. Therefore, successfully applying EPs demands tailored strategies that address each project's unique context and align project goals with stakeholder expectations. This includes developing a robust framework for monitoring and evaluation that navigates the intricacies of diverse stakeholder interests and external limitations on capacities and resources.

Table 3
Potential actor challenges to integrating more robust social considerations into AR4D systems.

Potential AR4D System Actor Challenges	
Complex power dynamics	Navigating systemic biases
Geographical and representational disconnect	Compartmentalizing structural challenges
Unintended marginalization	Lack of diverse perspectives
Integration of social considerations	

Co-creation can be problematic without structure: While advocating for co-creation in innovation scaling, there are challenges to moderating and managing the numerous stakeholders (Woltering et al., 2019). For example, engaging stakeholders without a structure and strategic thinking may risk wasting stakeholders' time and thus creating distrust and resentment in the innovation process. Additionally, stakeholders may include the potential "winners" and "losers" of innovation scaling. Without the ability to moderate the dynamics and understand actors' underlying interests, facilitators can easily be drawn into the conflicts of stakeholders. Furthermore, cross-learning is not a straightforward process. It can be difficult for some stakeholders to listen to, accept, and support the perspectives of others. Everyone has their own biases, discrimination, and different levels of openness (Hubbard, 2014; World Bank, 2014). Facilitating multi-stakeholder platforms and dialogue and working with partners that have established trust among stakeholders can help manage this and mitigate potential conflicts (Ratner et al., 2018; Sanginga et al., 2007).

Lack of synergized, accessible, and applicable tools: There are tools to support the operationalization of the EPs. However, they are often scattered and remain unavailable in local languages. Furthermore, many tools are extensive and time-consuming to learn and apply. Meanwhile, there is often a lack of facilitators who can understand and transfer the concepts in an accessible language to other stakeholders, including the end-users. The use of "Westernized" languages can also be a challenge to understand in local cultural context, e.g., interpreting words such as "transformation" and "innovation". There remains a lack of innovative ways to communicate EPs to enhance understanding and sympathy. Such innovative ways must help actors relate to their environment and construct actionable solutions.

Path dependency: Path dependency and technological lock-in resists systems and actors to make transformative changes. Such path dependency may exist in all actors and systems (Conti et al., 2021). Efforts supporting the operationalization of EPs must realize this challenge and support leveraging actions to break such path dependencies.

Causal processes creating inclusive and equitable innovation: We have been unable to analyze how EPs create causal impact pathways on inclusive and equitable innovation. This limitation creates a barrier to understanding the critical points that trigger the enhancement of the inclusive and equitable innovation process. For example, in a middle-range causal inclusive innovation model, Hoffecker (2021) noted that triggering effective dialogue between stakeholders does not depend on the selection of project activities (e.g., co-design, scenario development, and field trial) but on how these activities are implemented to foster collective cognition and consensus formation. These insights are crucial in supporting inclusive and equitable innovation development, yet they are scarce in the literature.

Logistics and Resources: It is widely recognized that operationalizing AIS frameworks is challenging due to the complexity of the AR4D system, which inherently involves biases and priority-setting. The difficulties of operationalizing AIS frameworks are compounded by the involvement of multiple actors, landscape-level effects, and downstream unknowns, particularly under evolving systems. Furthermore, AR4D interventions are often limited in scope, primarily due to government and funder responsibilities to the public, resulting in short project cycles with limited activities. As such, transformative work involving reflexive and participatory problem-solving processes, or the development of long-term education and governance programs is often not accommodated. Donors play a key role in this ecosystem, increasingly supporting the application of EPs, whether such principles are mentioned explicitly

or implicitly. Their advocacy and financial backing are instrumental in propelling the adoption of such principles through AR4D activities. Implementing EPs require flexible visioning processes, re-prioritization of existing resources, and/or an increase in overall resources for this purpose.

5.2. Perspective for future studies

For over 60 years, various approaches to promoting innovation in agricultural systems have emerged and evolved. Each approach responds to the paradigms and structures of their time to understand and improve agricultural systems. Klerkx et al. (2012) recognized that AR4D actors have different approaches that coexist and, in some cases, nurture each other. We appreciate the advancements within AIS and AR4D tools aimed at integrating AIS frameworks to facilitate more effective research, innovation, and scaling. Building on this progress, we believe that in the new era of AIS understanding, exploring social differentiation in both upstream and downstream processes and their effects represents a crucial area for further exploration.

While the EPs aim to support the integration of social differentiation in AIS, a comprehensive impact assessment of the proposed principles has yet to be realized. Substantial investments, particularly in the formative stages, are imperative, focusing on cultivating soft skills to operationalize the EPs. However, the long-term implications of such investments and their impacts remain unknown. Consequently, comprehensive assessments are warranted to enhance our comprehension of the business rationale for applying the EPs. These assessments should aim to identify the decisions necessary to support social inclusion from the donor's perspective, characterize the requisite tools, and explore potential synergies in operationalizing the EPs. Appraisals should also delineate the impact pathways associated with using the EPs, quantifying leverage points and overall impacts. Multiple research methodologies are essential for capturing these decisions and impacts effectively.

5.3. Conclusions

The EPs offer widespread applicability across various contexts. Initiatives like the Responsible Research and Innovation (RRI) framework (European Commission, 2013) and the CGIAR'S Independent Science for Development Council (ISDC) increasing interest in transformation through inclusive Innovation, underscore the growing recognition and commitment toward inclusivity in innovation (Independent Science for Development Council, 2022). Similarly, USAID has developed strategies that align with the EPs' values, aiming to foster equitable local partnerships and emphasizing commitments to gender, youth, and other marginalized groups (USAID, 2022). The World Bank has also recognized the importance of integrating social considerations of gender and equity within their AIS investment guidelines (World Bank, 2012), further emphasizing the longstanding, and growing upstream commitment to these ideals.

Moreover, the EPs present an opportunity to engage new user groups in agricultural innovation, enabling AR4D actors to better anticipate and mitigate the unintended consequences of innovation and scaling, thus fostering positive social impacts. By introducing the EPs, the aim is to enhance the effectiveness of AR4D actors in their endeavors. As the global community moves toward the SDGs, the EPs are positioned as essential to AR4D efforts, providing a framework for integrating these principles deeply into AR4D activities to support the SDG 2030 agenda meaningfully.

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CRediT authorship contribution statement

Erin McGuire: Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Maha Al-Zu'bi:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Maria Boa-Alvarado:** Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis, Conceptualization. **Thi Thu Giang Luu:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Janelle M. Sylvester:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Eva Marina Valencia Leñero:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Equity analysis of the MLP framework and the scaling readiness and PLC tools

A.1. MLP framework

Summary: The MLP is a comprehensive framework that encompasses various socio-technical aspects of the system and is suitable for application in the agri-food sector. This socio-technical system approach considers three elements as part of the system: landscapes (macro-level characteristics of a system), regimes (structures, institutions of a time and place), and niches (the individuals developing new ideas) (Geels, 2011). According to this framework, a system transformation change results from the interactions and alignment of multiple transition pathways through time (Geels and Schot, 2007).

Strengths and opportunities: Unlike other AIS frameworks, the MLP provides a prescriptive approach to sustainability transitions from a complex social change perspective, which is not focused on the agri-food sector. However, given the current trend of agri-food systems toward sustainability, the MLP framework can be used as an agricultural innovation system. It describes a process that involves the acceleration or "scaling" of innovations (niche) to build them up as part of structural normality (regime). The framework acknowledges that the world's landscape is beyond our control and that innovation processes may take much longer. The framework proposes steps required to scale an innovation in a system.

Equity analysis: The MLP is based on a socio-technical systems approach that blends technological, sociological, and institutional theories, but it has been criticized for its treatment of agency (Geels, 2011; Geels and Schot, 2007). The MLP underpins the role of agency in transitions but does not address power and political relationships (Genus and Coles, 2008; Smith et al., 2005). Furthermore, applying these conceptual levels empirically poses a challenge in terms of clarity, making it difficult to apply possible insights that result in action in practice (Smith et al., 2005). Geels has reflected on both of these critiques and acknowledged that the MLP does not yet differentiate between global and local agency levels, nor does it describe some agency types related to power, civil society, and cultural dimensions (Geels, 2011). In response to these critiques, the MLP could define better how to scope the system that it analyzes and better describe the agents involved in a transition from a power and/or cultural relationship perspective.

Entry points for social differentiation: An improved social differentiation analysis could benefit the MLP in two ways. Firstly, the scope of systems needs to be refined from a global to a local scale. This will provide actionable insights for stakeholders to drive sustainability transitions (Hölscher et al., 2023). Essentially, this involves translating the dimensions of the MLP into a geographically based system, identifying specific agents at each level. Secondly, the MLP should incorporate an analysis of agents or stakeholders, focusing on their power dynamics and cultural backgrounds. Instead of merely mapping these agents geographically, it is crucial to understand their roles and influences from both a historical and cultural standpoint, emphasizing their power or cultural attributes. An understanding of who these agents are would bring information on their behaviors, intentions, interrelations, agendas, interests, and influences (Brugha and Varvasovszky, 2000), which would then tell us more about the dynamics of this process and what is feasible from an agent perspective.

Emerging gaps

- **Power dynamics and actor agency**: The MLP Framework does not sufficiently address how power and political relationships influence transition processes. Additionally, it lacks agency related to global and local, civil society, and cultural dimensions adequately.
- **Contextual operationalization**: The MLP faces challenges in providing clear operational guidelines adaptable to various socio-cultural and geographical contexts.
- **Data availability and quality**: High-quality, context-specific data is crucial for identifying promising niches and understanding the intricacies of existing regimes.
- **Expertise and interdisciplinary collaboration**: The MLP's emphasis on socio-technical transitions necessitates a broad range of expertise spanning technical, social, and environmental domains to navigate the complexities of changing systems.

Relevant social theories: Power Theory; Feminist Theory, Human Rights Theory, and Social Network Theory.

Resource requirements and operationalization challenges: To address the criticism of how to scope the systems better from the global to local spheres, Geels himself has clearly addressed this question (Geels, 2011). He proposed that for the operationalization, there should be first a demarcation of the object of the analysis and then the operationalization of the regime in the MLP. He stressed that the MLP can adapt to different scopes of analysis, depending on the object of study. Therefore, to apply it to the agri-food system, the scope in the agri-food systems would depend on the level of complexity required. The limitation of the scope could be done in different ways including a geographical (global to local) or societal scope (individual, firm, industry, societal).

Different sectors' power and cultural perspectives have not yet been applied to the MLP Framework despite new insights from different disciplines (Geels, 2011). The analysis of stakeholders from a power perspective could benefit from information on what types of stakeholders and roles are required in innovation systems for transformation from a demand and supply perspective. From a supply approach, the sustainability process in the MLP can be described by the relationships between four types of actors: frontrunners, connectors, topplers, and supporters (de Haan and Rotmans, 2018). The actors from the demand level and their relationships can be analyzed from the types of actors from the diffusion of innovation described by Rogers (1962): innovators, early adopters, early majority, late majority, and laggards. This approach provides a comprehensive view of innovation's supply and demand dynamics. Such understanding allows for a detailed examination of existing relationships and identification of potential gaps within an (agri-food) value chain.

Furthermore, from a cultural perspective, the MLP could analyze who these stakeholders could be from an intersectional approach. For this, the diversity of the stakeholders involved in a transition could be analyzed using the diversity circle (Fig. A1; Hawkins et al., 2017), or an analysis could identify the groups vulnerable to discrimination in the transition process. For the latter, one could analyze vulnerable groups based on five variables that consider what defines a vulnerable group: place of residence, socio-economic status, governance, and vulnerability to shocks (UN, 2021). The degree of vulnerability would depend on the context.

Fig. A1. Diversity Circle (Hawkins et al., 2017).

A.2. Product life cycle tool

Summary: The PLC approach focuses on the different stages of an agricultural product's life, from research and development to its introduction, growth, maturity, and eventual decline. The Product Life Cycle (PLC) represents the latest enhancement to USAID's suite of scaling tools, augmenting the Agriculture Scalability Assessment Tool (Kohl and Foy, 2018) and serving as a critical component of a decision-making tree designed to aid innovation managers with scaling decisions (Kohl and Foy, 2018, p. 49). This approach emphasizes the need for continuous innovation and adaptation to changing landscape conditions, consumer preferences, and environmental factors to remain competitive and sustainable. The PLC aims to facilitate innovation adoption by ensuring a good fit between product and market and coordination between various actors.

Strengths and opportunities: The PLC approach is a tool used by USAID to guide the design, development, and delivery of innovation (predominately aligned with products) that could significantly impact development outcomes. The PLC approach recognizes that products go through a series of stages, from ideation to retirement, and that the appropriate strategies and resources required to move the product through each stage will vary. The approach also incorporates stage-gating to help limit the advancement of inappropriate innovations for end-users. It is an end-to-end management tool that advances products based on fitness and identifies downstream actors who must be engaged at each stage as a product advances. The PLC encourages continuous innovation and adaptation, considering the dynamic nature of markets and consumer preferences.

Equity analysis: The PLC process subtly engages various actors and networks yet lacks transparency in delineating ownership and leadership, obscuring the diversity among decision-makers. The PLC's approach, primarily designed for technological innovations, struggles to adapt seamlessly to social and institutional innovations, despite their inclusion in current practices. Many innovations necessitate complementary innovations or "innovation packages" for effective use, tailored to different user groups. Although the PLC acknowledges the role of complementary innovations, it does not place a strong emphasis on their evaluation. Moreover, the PLC's oversight of alternative scaling approaches, which might better serve marginalized groups through public sector and grassroots initiatives, underscores a potential bias toward methodologies and strategies commonly adopted in high-income marketplaces. Such a bias threatens to diminish the autonomy, expertise, and traditional knowledge of local and Indigenous communities, thereby reinforcing longstanding power disparities. Without a concerted effort to rectify these imbalances, there's a risk that dominant groups will consolidate control over benefits and decision-making processes, exacerbating the exclusion of less empowered stakeholders. Further, in its current version there is no explicit or requited end-user base assessment. It is important to note that the PLC is a new tool, and USAID has intentions of incorporating aspects of "*Scaling for Widespread Adoption of Improved Technologies and Practices*" which speaks to actor agency, public-private scaling pathways, gender inclusion, and more landscape assessments (USAID, 2023).

Entry points for social differentiation: The PLC stages provide opportunities to involve diverse stakeholders, to ensure marginalized voices are considered throughout. As such, this framework can help identify stakeholder involvement at each development stage, revealing actors' existing or missing capabilities at each step. Adaptations to the PLC could facilitate analysis of other more organizational or social innovations. Finally, monitoring reflective assessments before, during, and after the project could encourage trade-off reflection and reactions. Additionally, introducing a mechanism to gauge how scaling affects different user groups could enhance the PLC's social impact.

Emerging gaps:

- **Power dynamics:** lacks comprehensive guidance on addressing power dynamics in upstream research processes or downstream complex contextual challenges.
- **Innovation relevance and contextualization:** In the early stages of innovation development (e.g., the "research and development" and "introduction" stages), it is crucial to understand the target community's needs and context to ensure the innovation's relevance and success. The PLC's emphasis on private-sector mechanisms might neglect the cultural and institutional norms that shape social behavior, limiting its effectiveness in incorporating local realities.

- **Inclusiveness and responsiveness:** The discretionary inclusion of tools for analyzing power dynamics and social inclusion may lead to ineffective strategies for overcoming barriers for certain groups.
- **Incomplete guidance on considering social components:** Stage-gates do not incorporate social components or reflections on unintended consequences.
- **Social inequity reinforcement:** The framework may reinforce social inequities by privileging innovations developed within certain paradigms, potentially sidelining marginalized populations.

Gaps derived from the following social theories: Institutional Theory, Power Theory, Feminist Theory, Decolonial Theory.

Resource requirements and operationalization challenges: Social inclusion tools that are not synergized and documented introduce barriers to applying relevant tools in PLC. In cases where tools are developed, building capacity for facilitators who can effectively integrate social differentiation into innovation processes (e.g., integrating gender in innovation design, delivery, and impact assessment) takes time, especially in scaling contexts. Often, key facilitators are expected to possess 'transformed mindsets' regarding social inclusion. However, this capacity building or change process takes time, and only a limited number of individuals can fulfill this expectation. Additionally, AR4D projects might overburden participants and local facilitators if numerous surveys, orientations, and discussions are conducted for awareness-raising and data collection during ideation, M&E, and impact assessment. Another challenge arises from funders. While ensuring equity in innovation uptake and impacts among different social groups is a complex process, there is still a dominant use of a deterministic and linear approach (e.g., fixed indicators, short timelines, fixed finance) for financing innovation (Benton and Bailey, 2019). This approach risks only reaching the 'low-hanging fruits'—i.e., the easily accessible farmers with more power than others. To achieve positive social outcomes, it is important to reach and benefit an extended group of farmers and their households and communities.

A.3. Scaling readiness tool

Summary: The CGIAR's Scaling Readiness approach is a framework that aims to assist AR4D (organizations, programs, and projects) interventions in developing, implementing, and monitoring the scaling of innovations in a structured and evidence-based way. It assesses an innovation's readiness or maturity for scaling and the actions that could improve or accelerate the process. Core concepts include limiting factors to scaling, innovations scale with other innovations (e.g., "core" and "complementary"), and that successful scaling is a function of innovation readiness and innovation use (Sartas et al., 2020). Therefore, the unit of analysis for assessing scaling readiness is the innovation package as a whole. The following steps comprise the Scaling Readiness framework:

- Step 0: Fit-for-purpose. Determine that the use of the tool is benefiting the research project with adequate resources and time for implementation.
- Step 1: Characterize. Understand the innovation, core and complementary components, context, and stakeholders involved.
- Step 2: Innovation Profile Development. Diagnose and evaluate the scaling readiness and current use of each innovation in the package.
- Step 3: Innovation Packages. Strategize and analyze the strategic options to increase the scaling readiness with stakeholder analysis for potential solutions and partnerships.
- Step 4: Agree. Seek agreement among stakeholders on scaling investment decisions and facilitate learning and negotiation.
- Step 5: Scaling Strategy. Identify emerging windows of opportunity, tipping points, and the process of monitoring, evaluation, and learning.

This framework examines factors such as the innovation's effectiveness, feasibility, acceptability, and potential for impact at different scales. The Scaling Readiness framework was developed in 2018 as a conceptual framework (Sartas et al., 2020) and decision-support process (Schut et al., 2022) and has since been in continuous development as a tool that AR4D actors can use to understand how to scale their innovations. This is usually done over a one- to three-day workshop with a trained facilitator. Currently, Scaling Readiness is being transformed to be used throughout the CGIAR as an innovation management and curation database, referred to as the Innovation Packages and Scaling Readiness (IPSR). This analysis focuses on the Scaling Readiness concepts.

Strengths and opportunities: The Scaling Readiness framework is a rigorous, systematic approach that allows for comparative analysis and the consolidation of insights across projects (Sartas et al., 2020). It supports the performance management of the portfolio of innovations through time, by having a standardized and visual measurement for each core and complementary innovation(s). Scaling Readiness is a tool that analyzes the innovations' maturity and can provide information on how to improve these innovations over time and inform the interested stakeholders of the innovation's opportunities and challenges. Scaling Readiness is also advantageous in that it supports the development of innovation packages for different scaling contexts, allowing scaling strategies to be assessed with unique characteristics. This approach emphasizes iterative and adaptive management, encouraging periodic diagnoses throughout the scaling process, not only at the beginning of the project (Schut et al., 2022).

Equity analysis: Scaling Readiness includes tools specifically designed to analyze and facilitate the scaling of innovations. Thus, some of the strengths of Scaling Readiness from a technological perspective are also part of its critiques from a responsible innovation perspective. As the framework focuses on the innovation itself, it does not yet offer comprehensive guidance to reflect and decide upon certain social components during the innovation's scaling development phases. These include understanding power dynamics, considering complex contextual challenges when developing an innovation, and reflecting on unintended consequences and outcomes of the scaling of the innovation. In addition, it does not explicitly address gender and other vulnerable groups and minorities' needs as part of the selection and monitoring of the evaluation. This last point provided the impetus for creating the GenderUp Tool, which was developed to be used independently or jointly with Scaling Readiness to foster gender-responsible scaling (McGuire et al., 2023).

While complementary analytical tools can be used to understand power dynamics (e.g., social network analysis, power mapping, intersectional gender assessment) in the phases in which stakeholders are identified and engaged (e.g., Steps 1, 3, 4), the use of these tools is at the discretion of the scaling team. Scaling Readiness, as stated by the authors in Sartas et al. (2020), by design does not gauge the inclusiveness and responsibility of an innovation's scaling. However, through the process of including this tool as part of a larger initiative, the CGIAR's Innovation Platform for Scaling Readiness, there has been some work to analyze alternatives to include more gender and other stakeholder engagement systematically. For example, there is now a training and a manual for trainers that will use Scaling Readiness that recommends using participatory tools, and two questions have been included to increase gender reflection through the process. Although Scaling Readiness provides entry points for social inclusion, it allows upstream actors the discretion to analyze authority structures and biases or to use inclusive tools like social network mapping. This flexibility can

hinder the design of effective strategies to address scaling barriers for certain groups of people, such as unequal access to resources. Not making these tools integral to the process downplays their importance in driving social transformation.

Entry points for social differentiation: Beyond understanding the technological process when scaling an innovation, other elements that assess the readiness of an innovation in a context could be incorporated to consider social differentiation. Addressing power dynamics in the agricultural sector is a complex challenge. Nonetheless, enhancing the framework's focus on discerning and analyzing power dynamics (considering both top-down and bottom-up approaches), acknowledging trade-offs, developing safeguards, and recognizing intersectional vulnerabilities would be critical to increasing this tool's impact for effective social transformation. Additionally, the tool could offer broader prompts emphasizing market-oriented innovations and recognizing local knowledge and practices with scaling potential. By focusing on the target landscape and its interactions with innovation processes, Scaling Readiness can better promote the integration of social theory for effective, contextual scaling.

The different steps within the Scaling Readiness framework present various opportunities to consider social differentiation. Step 1 of the analysis, when characterizing the stakeholders, their networks, and their interventions, presents a critical opportunity to understand the users of the innovation (including women, youth, and other vulnerable groups in societies) and their needs. This assessment could go beyond the understanding of the technical capacities to comprehend the current mindsets and capacities of the people that are targeted to adopt the innovation in the future and develop relationships and connections with them to understand their innovation concerns. If necessary, capacity-strengthening strategies could be included as part of the scaling process that Scaling Readiness is driving to bring more people in as part of the process. The training could be for innovators, as is currently being developed, and other external agents that can be more neutral to the power dynamics.

Emerging gaps:

- **Power dynamics:** There is a lack of comprehensive guidance on addressing power dynamics in upstream research processes or complex downstream contextual challenges.
- **Innovation relevance and contextualization:** This is essential for the "Characterize" and "Innovation Profile Development" steps, where aligning the innovation with the context and needs of the stakeholders is crucial to ensure the effectiveness and acceptability of the scaling process.
- **Inclusiveness and responsiveness:** The discretionary inclusion of tools for analyzing power dynamics and social inclusion may lead to ineffective strategies for overcoming barriers for certain groups.
- **Incomplete guidance on considering social components:** The focus on innovation's technical maturity overlooks critical social components, such as power dynamics and the needs of vulnerable groups during the scaling process or reflection on unintended consequences.
- **Social inequity reinforcement:** The framework may reinforce social inequities by privileging innovations developed within certain paradigms, potentially sidelining marginalized populations.

Relevant social theories: Institutional Theory, Power Theory, Feminist Theory, Decolonial Theory, Decentralization Theory, and Social Capital Theory.

Resource requirements and operationalization challenges: Conducting social analyses with an intersectional lens or mapping power dynamics requires resource-intensive activities, such as conducting in-situ interviews and focus groups. These activities are sometimes logistically challenging, particularly in conflict-affected areas. Collecting comprehensive and representative data can also require ethics approval and proper data management. Gaining experience with social science methods and multidisciplinary teams is also a resource-heavy activity.

There are often insufficient data and resources on the innovation's characteristics and development to analyze it from an intersectional perspective that considers power dynamics. There are insufficient accurate data from interventions or projects to objectively identify bottlenecks in a given context, which hinders evidence-based decision-making for innovation development. Insufficient financial and human resource investments focus on social inclusion and responsible approaches that would allow adapting the current framework to new audiences and needs. Operational challenges would need to be addressed even if there were enough resources and data to drive responsible discussions forward. These include a) balancing innovation design goals and context's needs, b) designing for long-term sustainability and outcomes subject to real and changing world conditions, c) acknowledging which power dynamics are within reach of the project manager's control and which are not, d) learning about which efforts to include diverse representation can translate into results according to the cultural characteristics of a context, and e) including transparency about the innovator's and users' expectations and roles.

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